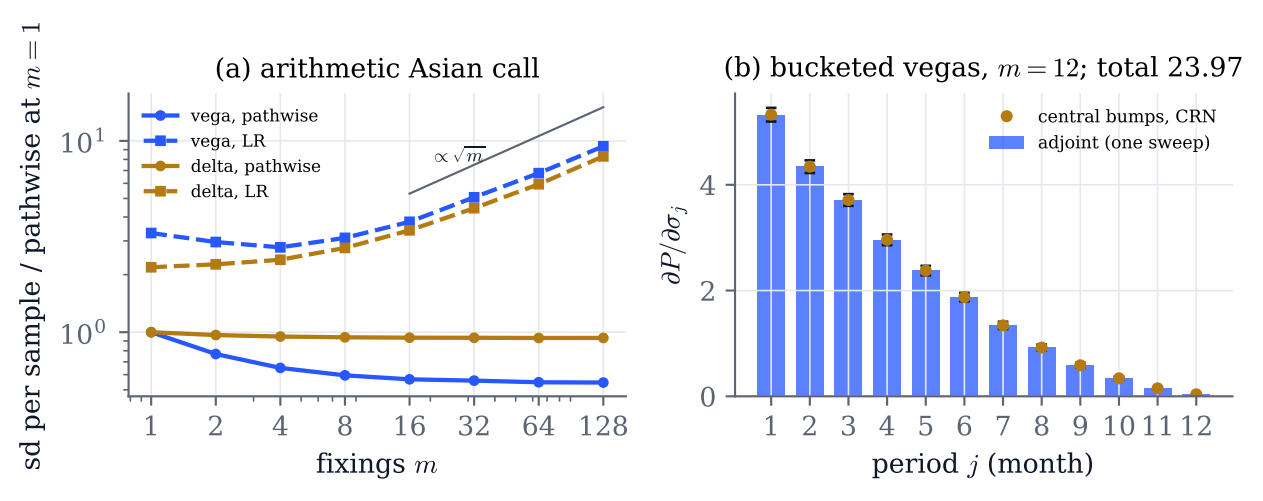
(a) arithmetic Asian call
vega, pathwise
vega, LR
delta, pathwise
delta, LR
$\propto \sqrt{m}$
10^1
10^0
sd per sample / pathwise at $m = 1$
1
2
4
8
16
32
64
128
fixings m
(b) bucketed vegas, $m = 12$; total 23.97
central bumps, CRN
adjoint (one sweep)
$\partial P / \partial \sigma_j$
0
2
4
1
2
3
4
5
6
7
8
9
10
11
12
period j (month)